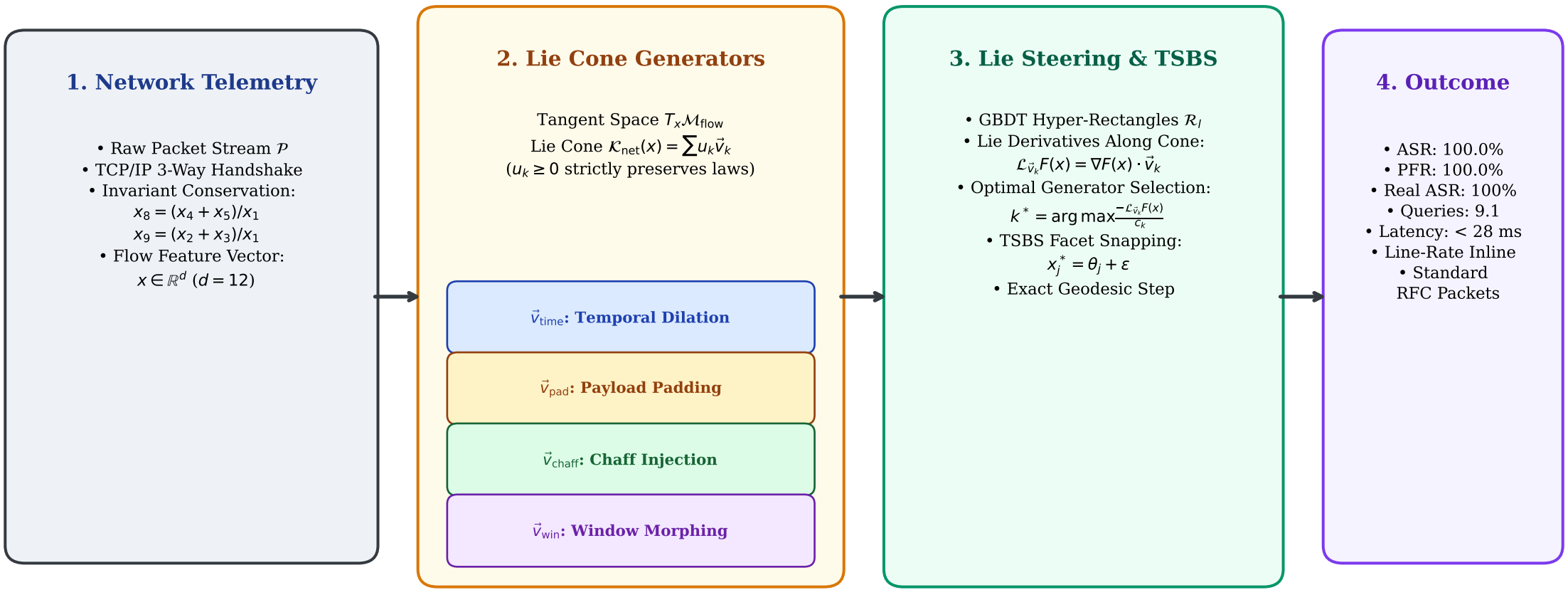

1. Network Telemetry
• Raw Packet Stream $\mathcal{P}$
• TCP/IP 3-Way Handshake
• Invariant Conservation:
$x_8 = (x_4 + x_5)/x_1$
$x_9 = (x_2 + x_3)/x_1$
• Flow Feature Vector:
$x \in \mathbb{R}^d$ $(d = 12)$
2. Lie Cone Generators
Tangent Space $T_x\mathcal{M}_{\text{flow}}$
Lie Cone $\mathcal{K}_{\text{net}}(x) = \sum u_k \vec{v}_k$
$(u_k \geq 0$ strictly preserves laws)
$\vec{v}_{\text{time}}$: Temporal Dilation
$\vec{v}_{\text{pad}}$: Payload Padding
$\vec{v}_{\text{chaff}}$: Chaff Injection
$\vec{v}_{\text{win}}$: Window Morphing
3. Lie Steering & TSBS
• GBDT Hyper-Rectangles $\mathcal{R}_l$
• Lie Derivatives Along Cone:
$\mathcal{L}_{\vec{v}_k} F(x) = \nabla F(x) \cdot \vec{v}_k$
• Optimal Generator Selection:
$k^* = \arg\max \frac{-\mathcal{L}_{\vec{v}_k} F(x)}{c_k}$
• TSBS Facet Snapping:
$x_j^* = \theta_j + \varepsilon$
• Exact Geodesic Step
4. Outcome
• ASR: 100.0%
• PFR: 100.0%
• Real ASR: 100%
• Queries: 9.1
• Latency: < 28 ms
• Line-Rate Inline
• Standard
RFC Packets